Estimating latent factors in dynamic models: the Kalman filter

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Abstract

In a 1988 working paper, Stock and Watson proposed a framework to estimate and forecast short-run U.S. economic activity. Their model built on the premise that comovements across multiple macroeconomic variables can be captured by a single, unobserved latent variable. Specifically, they hypothesized that their four chosen indicators form a dynamic one-factor model, where the latent factor represents the state of the business cycle and can therefore be mapped to Gross Domestic Product (GDP) growth (Stock and Watson 1988).

The objective of this project is to empirically evaluate and extend the Stock and Watson (S&W) model. First, we will verify the assumptions of one-factor models, specifically by testing the tetrad constraints in a dynamic settings. Then, we will asses its predictive performance by replicating the model on more recent data and comparing the estimated latent factor to actual GDP growth. Furthermore, because macroeconomic data frequently exhibits heavy tailed behaviors, we will test the robustness of the Kalman filter by simulating economies under progressively more elliptical distribution.

Finally, we will expand the number of indicators included in the model by incorporating more macroeconomic variables. To avoid adding too many variables, and therefore too much noise to the model, we will apply Sparse Principal Component Analysis (SPCA). Specifically, we will force sparsity on the new indicators, while simultaneously not assigning any penalty to the original four indicators in order to retain them. With this setup, we will ensure that the original four indicators will still represent the foundation of our model, while still squeezing extra predictive value out of a larger dataset.

1. The Stock & Watson model

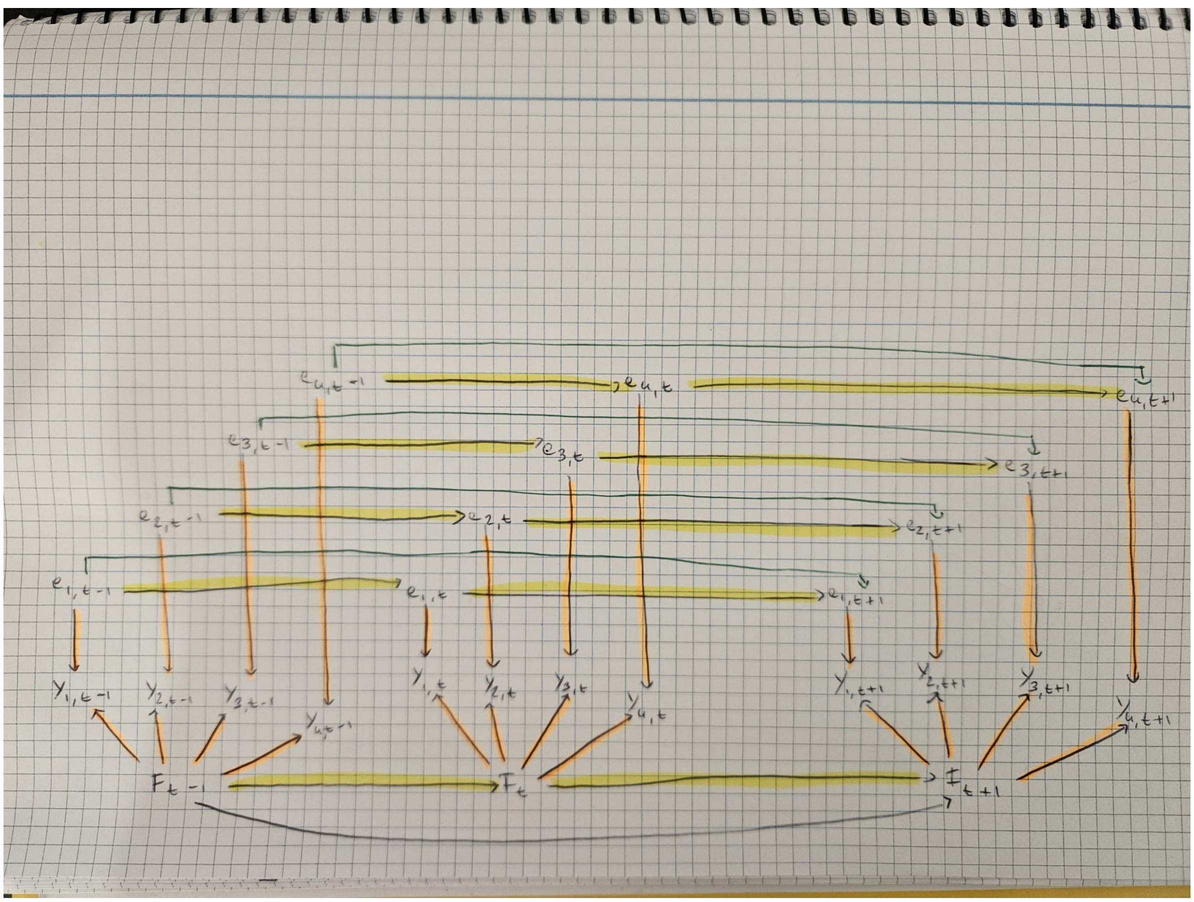
We first replicate model proposed by Stock and Watson. Their model assumes that the co-movements of four macroeconomic indicators are driven by a single, unobserved latent factor which represents the overall business cycle. Formally, the model is represented by the following structural equations:

- The measurement equation: $y_t = \lambda f_t + e_t$
- The factor transition equation: $f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + w_t, \quad w_t \sim i.i.d.N(0,1)$
- The error transition equation: $e_t = \psi_1 e_{t-1} + \psi_2 e_{t-2} + \epsilon_t, \quad \epsilon_t \sim i.i.d.N(0,\sigma_i^2)$

Where:

- y_t is the (4×1) vector of normalized log-differences of the indicators at time t
- λ is the (4×1) vector of coefficients
- f_t is the (1×1) factor scalar at time t
- e_t is the (4×1) vector of errors at time t

Noticeably, to capture the cyclic patter of business cycles, both the factor and the error follow an $AR(2)$ process. Visually, the model can be represented by the following graphical model:



1.1. Latent factor estimation

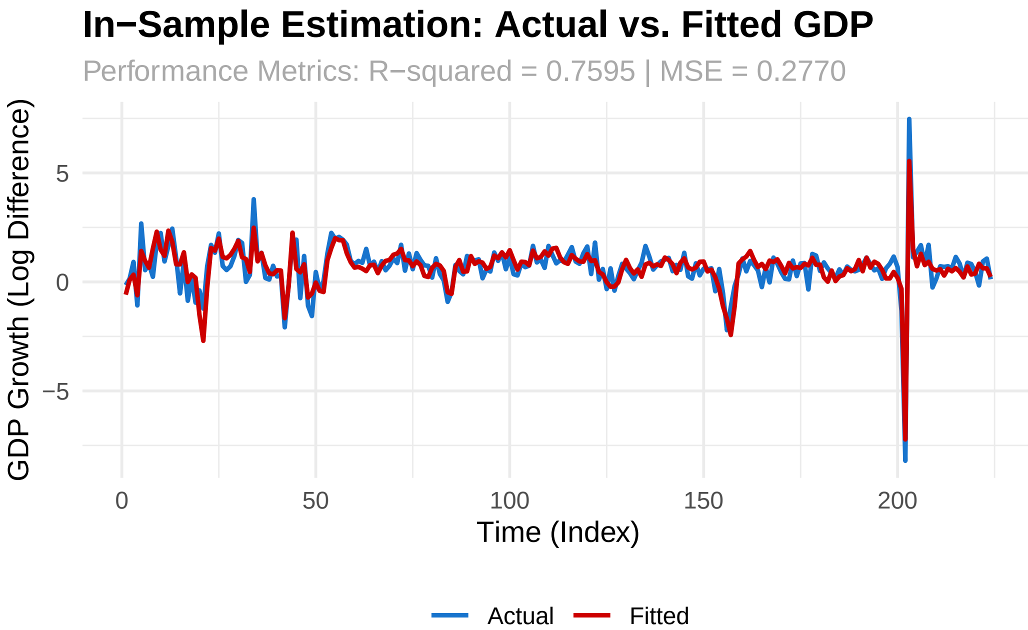
Given that the model is linear and that shocks w_t and w_t are assumed to be Gaussian and independent, the optimal estimator for the latent factor is given by the Kalman filter (Kalman 1960). However, this estimation technique requires the errors of the measurement equation e_t to be independent white noise, differently from the $AR(2)$ specification of the S&W model. To overcome this limitation, we take the $AR(2)$ errors out of the measurement equation and include them into the latent state vector, alongside the current and lagged values of the factor. The structural equations simplify to:

- The measurement equation: $y_t = Hh_t$
- The factor transition equation: $h_t = Fh_{t-1} + v_t, \quad v_t \sim i.i.d.N(0, \sigma)$

Where h_t is the state vector which now includes the current and lagged values of both the factor and the error. With this structure, the F matrix handles the $AR(2)$ dynamics of both the factor and the error.

1.2. Mapping the latent factor to GDP growth

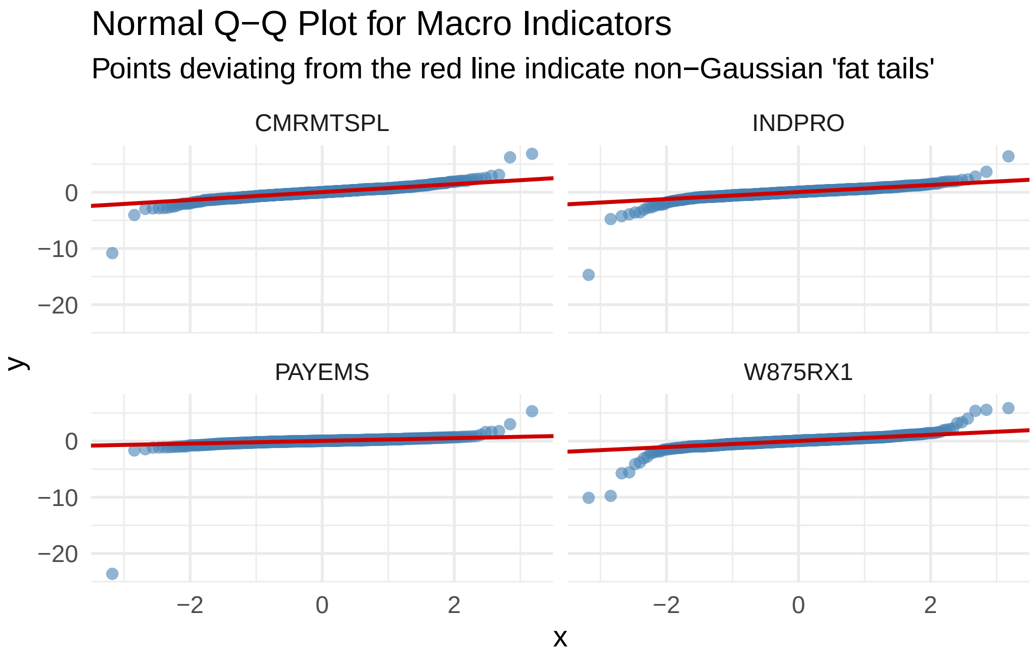
After estimating the latent factor via the Kalman filter, we must rescale it to make it interpretable in terms of actual economic activity. To do so, we regress GDP onto the estimated factor by Ordinary Least Squares (OLS). This allows us to plot the estimated latent factor against GDP growth and calculate in-sample performance metrics, such as the coefficient of determination (R^2) and the Mean Squared Error (MSE).



The empirical results demonstrate the strong explanatory power of the model proposed by Stock & Watson. The mapped factor explains approximately 76% of the total variance of the actual GDP growth ($R^2 = 0.7595$). Moreover, the model yields an in-sample MSE of 0.277, translating to a Root Mean Squared Error (RMSE) of roughly 0.526 percentage points. This RMSE might seems elevated, but it's important to note that our dataset includes periods of extreme macroeconomic volatility, such as the 2008 and COVID-19 crises, which inflate the error metrics.

1.3. The normality assumption

The dynamic factor model proposed by Stock and Watson explicitly assumes that all structural shocks are purely Gaussian. However, economic cycles frequently feature sudden and massive shocks (e.g. the COVID-19 crisis), hence macroeconomic indicators are rarely normally distributed. Instead, they tend to exhibit heavy-tailed behavior. As illustrated in the Quantile-Quantile (Q-Q) plot below, the distribution of our indicators indeed deviates from normality, confirming the presence of fat tails.



Despite the violation of the normality assumption, the Kalman filter extracts a consistent and unbiased estimate of the latent factor. Indeed, when the parameters are estimated via Gaussian quasi-maximum likelihood, consistency of the parameter estimates is preserved under non-Gaussianity, though with loss of efficiency in finite samples as the distribution moves away from Gaussianity and gets increasingly elliptical (White 1982). Chapter 2 will be dedicated to an empirical demonstration of the robustness of the Kalman filter, by simulating the economic model proposed by S&W under increasingly elliptical distributions.

1.4 Verifying that there is only one underlying factor (TO FINISH)

However, from our class we know that tetrad constraints to verify that there is only one underlying factor still hold under more elliptical distributions. Note, however, that tetrad constraints cannot be identified in a static way anymore, because the one factor model is dynamic. However, we can still verify them on the autocovariance structure (expand the explanation).

Table 1: Empirical Dynamic Tetrad Constraints

Lag	Tetrad 1	Tetrad 2	Tetrad 3
Lag 0	0.0524	0.0732	0.0208
Lag 1	0.0414	0.0770	0.0356
Lag 2	-0.0116	-0.0046	0.0070

Lag	Tetrad 1	Tetrad 2	Tetrad 3
Lag 3	-0.0054	-0.0031	0.0023
Lag 4	0.0066	0.0065	-0.0001
Lag 5	0.0005	0.0006	0.0001
Lag 6	0.0036	0.0041	0.0004

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